

NOTE

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1 Final Project

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- Solver
- Must support parallelization
- Must use GitHub for development

2 Lattice Boltzmann Method (LBM)

- Fluids can be imagined as consisting of a large number of small particles undergoing random motion.
- The exchange of momentum and energy is achieved via particle **streaming** and **collision**.
- The LBM simplifies Boltzmann's idea of gas dynamics by reducing the number of particles and confining them to the nodes of a lattice.
- In actual implementations, **streaming** and **collision** are computed separately.

The process can be modelled by the Boltzmann transport equation

$$\frac{\partial f(\vec{x}, t)}{\partial t} + \vec{u} \cdot \nabla f(\vec{x}, t) = \Omega,$$

where

- $f(\vec{x}, t)$ is the particle distribution function.
- $\vec{u}$ is the particle velocity.
- Ω is the collision operator.

For a two-dimensional model, a particle is restricted to stream in 9 possible directions. A typical lattice node of the D2Q9 model with 9 velocities $\vec{e}_i$ is defined by

$$\vec{e}_i = \begin{cases} (0, 0) & \text{if } i = 0 \\ (1, 0), (0, 1), (-1, 0), (0, -1) & \text{if } i = 1, 2, 3, 4 \\ (1, 1), (-1, 1), (-1, -1), (1, -1) & \text{if } i = 5, 6, 7, 8 \end{cases}$$

We can associate a discrete particle distribution function with

$$f_i(\vec{x}, t), \quad i = 0, \dots, 8,$$

which describes the probability of **streaming** in one particular direction.

The *macroscopic fluid density* can be defined as the sum of microscopic particle distribution functions

$$\rho(\vec{x}, t) = \sum_{i=0}^8 f_i(\vec{x}, t).$$

The *macroscopic velocity* is the weighted (by the distribution functions f_i) average of microscopic velocities $\vec{e}_i$

$$\vec{u}(\vec{x}, t) = \frac{1}{\rho} \sum_{i=0}^8 c f_i \vec{e}_i.$$

The **streaming** and **collision** processes are given by

$$f_i(\vec{x} + c\vec{e}_i\Delta t, t + \Delta t) - f_i(\vec{x}, t) = -\frac{[f_i(\vec{x}, t) - f_i^{\text{eq}}(\vec{x}, t)]}{\tau},$$

where

- LHS represents **streaming**, and RHS describes **collision**.
- $f_i^{\text{eq}}(\vec{x}, t)$ is the equilibrium distribution.
- τ is the relaxation time towards local equilibrium.

The Bhatnagar-Gross-Krook (BGK) collision model is used to describe single-phase flows

$$f_i^{\text{eq}}(\vec{x}, t) = w_i \rho + \rho s_i [\vec{u}(\vec{x}, t)],$$

where $s_i(\vec{u})$ is defined as

$$s_i[\vec{u}(\vec{x}, t)] = w_i \left[3 \frac{\vec{e}_i \cdot \vec{u}}{c} + \frac{9}{2} \frac{(\vec{e}_i \cdot \vec{u})^2}{c^2} - \frac{3}{2} \frac{\vec{u} \cdot \vec{u}}{c^2} \right],$$

w_i are the weights

$$w_i = \begin{cases} \frac{4}{9} & \text{if } i = 0 \\ \frac{1}{9} & \text{if } i = 1, 2, 3, 4 \\ \frac{1}{36} & \text{if } i = 5, 6, 7, 8 \end{cases}$$

and $c = \frac{\Delta x}{\Delta t}$ is the lattice speed. The fluid kinematic viscosity ν in the D2Q9 model is related to the relaxation time τ by

$$\nu = \frac{2\tau - 1}{6} \frac{(\Delta x)^2}{\Delta t}.$$

The algorithm is summarized as

1. Initialize ρ , $\vec{u}$, f_i and f_i^{eq} .
2. Streaming: Move $f_i \rightarrow f_i^*$ along the direction of $\vec{e}_i$.
3. Compute macroscopic ρ and $\vec{u}$ from f_i .
4. Compute f_i^{eq} .
5. Collision: Calculate the updated distribution function f_i with

$$f_i = f_i^* - \frac{1}{\tau} (f_i^* - f_i^{\text{eq}}).$$

6. Repeat steps 2 to 5.

3 Boundary Conditions (BCs)

3.1 Bounce-back BCs

- When a fluid particle reaches a boundary node, the particle will scatter the fluid back along its incoming direction.
- There are two types of implementations: the on-grid and the mid-grid bounce-back.
 - The on-grid bounce-back is simple and computationally efficient, since this BC just reflects a fluid particle when it hits the wall.
 - The mid-grid bounce-back needs a fictitious node outside the wall to apply the reflection.
 - However, the on-grid bounce-back is only first-order accurate, and the mid-grid is second-order accurate.

3.2 Zou-He BCs

- The main idea is to model flows with given velocity, pressure, or density at the boundary.
- This type of BC depends on the orientation of the boundary and is thus hard to generalize for complex geometries.

Rewriting the *macroscopic fluid density* and *velocity* equations, we have

$$\begin{aligned}f_1 + f_5 + f_8 &= \rho - (f_0 + f_2 + f_4 + f_3 + f_6 + f_7); \\f_1 + f_5 + f_8 &= \rho u + (f_3 + f_6 + f_7); \\f_5 - f_8 &= \rho v - f_2 + f_4 - f_6 + f_7.\end{aligned}$$

ρ is thus

$$\rho = \frac{1}{1-u} [f_0 + f_2 + f_4 + 2(f_3 + f_6 + f_7)].$$

To solve for f_1 , f_5 , and f_8 , the assumption is that the bounce-back rule still holds for the non-equilibrium part of the particle distribution normal to the boundary. The fourth equation is given by

$$f_1 - f_1^{\text{eq}} = f_3 - f_3^{\text{eq}}.$$

Thus, f_1 , f_5 , and f_8 are

$$\begin{aligned}f_1 &= f_3 + \frac{2}{3}\rho v; \\f_5 &= f_7 - \frac{1}{2}(f_2 - f_4) + \frac{1}{6}\rho u + \frac{1}{2}\rho v; \\f_8 &= f_6 + \frac{1}{2}(f_2 - f_4) + \frac{1}{6}\rho u - \frac{1}{2}\rho v.\end{aligned}$$

4 Topics

- Steady Plane Poiseuille Flow
- Lid-driven Cavity Flow
- Flow past a Circular Cylinder
- Rayleigh-Bénard Convection

5 Connection with the Navier-Stokes Equations